TFScope: Learning the DNA-binding specificity of transcription factors from protein sequence

Lei Huang[1,2], Zain M. Patel[1,2], Luca Pinello[1,2]

[1]Molecular Pathology Unit, Krantz Family Center for Cancer Research, Massachusetts General Hospital / Harvard Medical School, Boston, MA, USA

[2]Broad Institute of MIT and Harvard, Cambridge, MA, USA

lhuang34@mgh.harvard.edu

Transcription factors (TFs) are central regulators of gene expression, controlling cell identity, and development by recognizing specific DNA sequences in the genome. A fundamental challenge in regulatory genomics and protein design is to decode how DNA-binding specificity is encoded in TF amino acid sequence without the structural information. Despite decades of experimental and computational studies, predicting TF–DNA binding preferences directly from protein sequence remains difficult, particularly across diverse TF families with distinct DNA-binding domain (DBD) architectures and recognition mechanisms.

Current approaches to TF specificity inference largely fall into three categories. First, homology-based methods transfer binding preferences between closely related TFs, but their applicability is limited in low-homology regimes and for poorly characterized families. Second, family-specific recognition codes—such as position-specific residue–base interaction rules—have provided valuable insights for selected TF classes (e.g., zinc fingers and homeodomains), yet these rules are often incomplete and do not generalize across DBD families with heterogeneous binding grammars. Third, experimental approaches such as SELEX, protein-binding microarrays, and high-throughput sequencing assays enable direct measurement of TF binding motifs, but are labor-intensive and not scalable to the full repertoire of TFs across species. There is a machine learning model which could predict the binding specificity, however, it requires the crystal structure of both protein and DNA sequences. As a result, there remains a critical gap in methods that can predict TF–DNA specificity from sequence in a generalizable, scalable, and mechanistically interpretable manner.

Recent advances in protein structure prediction, particularly with models such as AlphaFold 3, have opened new opportunities to study protein–DNA interactions at atomic resolution. In parallel, physics-based modeling frameworks such as Rosetta enable detailed estimation of binding energetics through in silico mutational scanning. However, purely structure-based pipelines remain computationally expensive and often lack efficient search strategies over the vast space of possible DNA sequences. Conversely, purely sequence-based machine learning approaches can scale but typically lack structural grounding and struggle to capture the biophysical determinants of specificity.

Here, we propose a framework **TFScope** that combines learned sequence priors with structure prediction and physics-based refinement to provide a principled route to bridge this gap. We introduce a structure-guided machine learning framework for de novo inference of TF–DNA binding specificity from protein sequence. At the core of our approach is a family-aware seed machine learning model that encodes TF sequence and DBD context to predict candidate motif lengths and coarse DNA-binding preferences. Specifically, the seed model is a family-aware neural architecture built on a pretrained protein language model ESM3, which encodes TF sequence into contextual residue representations and aggregates global and DNA-binding-domain-focused features. By explicitly conditioning on DNA-binding domain family identity, the model captures both shared and family-specific recognition logic, and jointly predicts candidate motif lengths and coarse nucleotide preference profiles to initialize downstream TF–DNA specificity inference by utilizing mix of expert (MOE) strategies. The predicted candidate motifs define a reduced search space of plausible DNA targets, which are subsequently instantiated as TF–DNA complex structures using AlphaFold3.

To achieve base-resolution specificity, we further integrate Rosetta-based in silico mutational scanning, systematically perturbing DNA sequences within modeled complexes to estimate position-specific binding energetics. These energetic profiles are then used to derive calibrated position weight matrices (PWMs), yielding a quantitative and mechanistically interpretable representation of TF–DNA binding specificity. In this way, our framework transforms TF sequence into a structure- and energy-informed specificity landscape, linking sequence, structure, and function within a unified pipeline. We would evaluate our approach using motif-level similarity metrics, base-resolution preference accuracy, and cross-family generalization benchmarks to demonstrate robust performance across diverse TF families.

Beyond motif prediction, this framework provides a versatile foundation for a range of applications. These include annotating DNA-binding specificity for TFs without recognized motifs, interpreting the functional impact of sequence variants on TF binding, and designing synthetic transcription factors with programmable DNA specificity. More broadly, our work illustrates how integrating machine learning with structural and energetic modeling can enable mechanistically grounded predictions for complex biomolecular interactions.

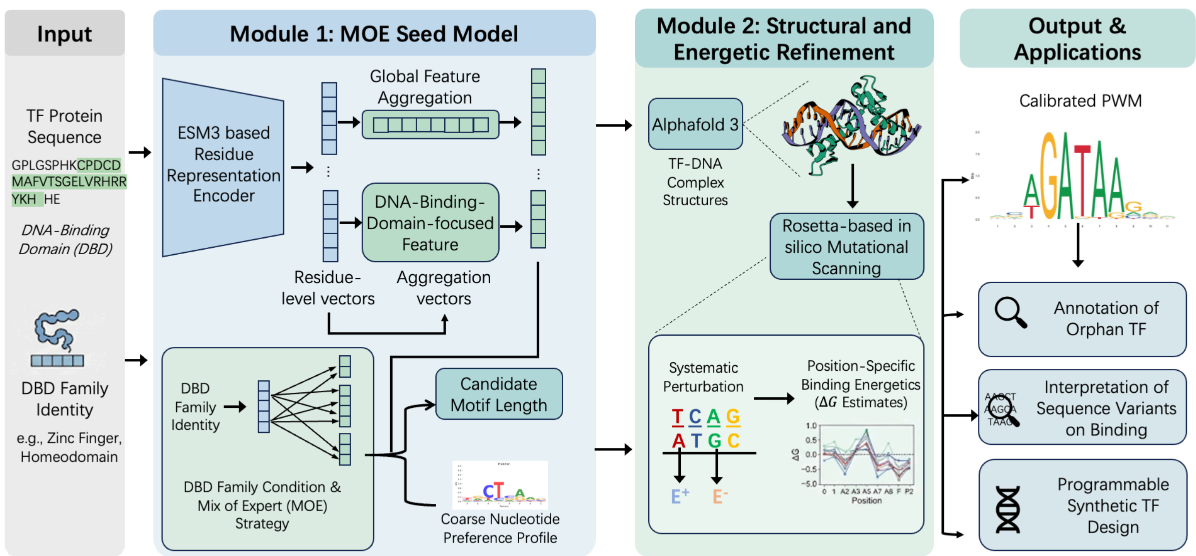

Figure 1. The diagram of TFScope framework

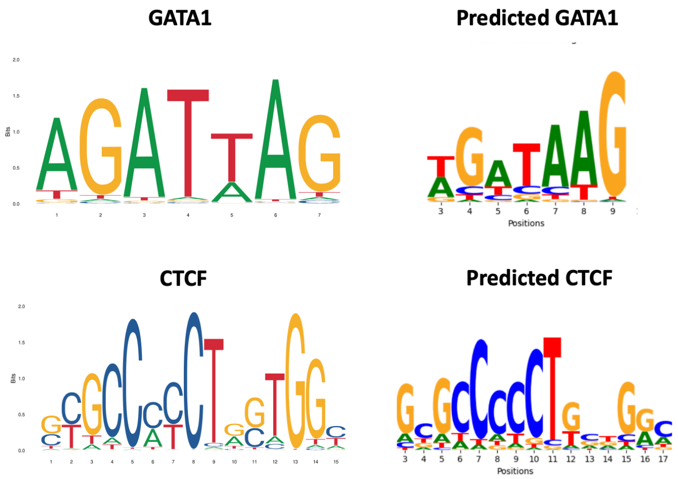

Figure 2. The predicted examples of TFScope